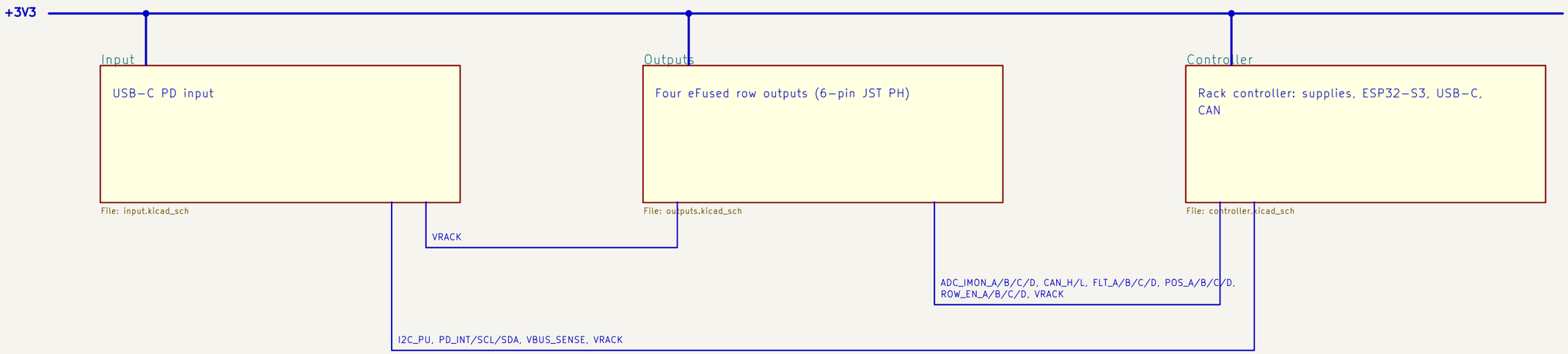


Power hub v0.5 — USB-C PD in (CH224A: 9 V, then 20 V or 28 V EPR as the controller asks) = VRACK → 4 × TPS26600 eFuse (1.6 A) → 6-pin JST PH rows A-D.
Rack controller: ESP32-S3-MINI-1 on VBUS, TJA1462AT (CAN master, star bus, both terminations here). POS token per row.
Rows are OFF without firmware: SHDN is pulled low; the controller turns each row on once the PD contract is final.

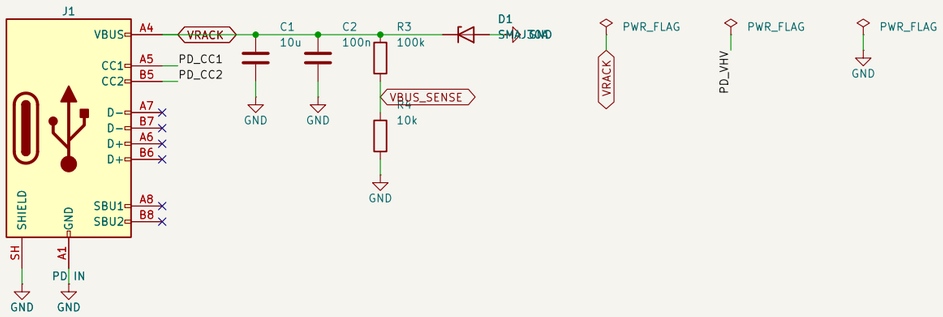
Generated from electronics/power-hub/design.py — edit the Python, then regenerate
USB-C PD distribution + rack controller (ESP32-S3, CAN master) for the hw-testing-rack
hw-testing-rack

Sheet: /
File: power-hub.kicad_sch

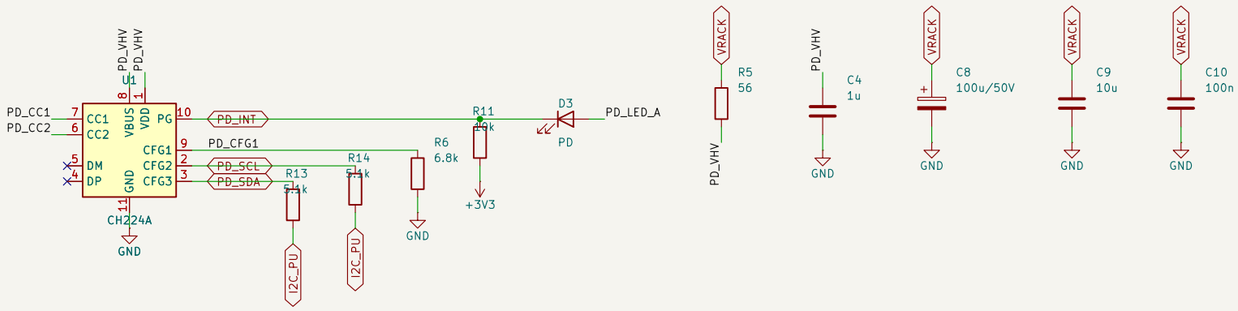
Title: Power hub — Overview

Size: A3	Date:	**Rev: 0.5**
KiCad E.D.A. 10.0.6		Id: 1/4

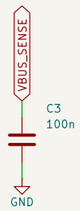
USB-C PD input (5 A receptacle, CC to the sink controller), TVS; VBUS is VRACK (no input switch)



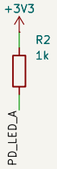
CH224A USB PD 3.2 sink: 9 V at plug-in (CFG1 6.8k), then 20 V / 28 V EPR as the controller asks over I2C



I2C to the controller, VBUS sense



PD status (front LED): on while the CH224A reports power good (PG low)



U1 is the CH224A (WCH, PD 3.2): CFG1 6.8k asks for 9 V at plug-in and turns its I2C on (0x22); the controller reads the source's PDOs (0x60-0x8F) and writes 0x0A: 5 = 28 V EPR when offered, else 4 = 20 V.
No input switch: VBUS is VRACK. The eFuses (UV 15 V, off until enabled, reverse current blocked) are the switches; the controller turns the rows on once 0x09 says PD (and EPR) and VBUS_SENSE reads the voltage.
SMAJ30A: 30 V stand-off, above a 28 V EPR contract; the eFuses (60 V) and the TPS26600 OVP (33 V) follow.
R5/C4: the CH224A's VHV/VBUS pins are rated 32 V; the RC filters the fast edges the TVS can't reach.
VBUS_SENSE: 100k/10k from VBUS → 2.5 V at 28 V for the ESP32's ADC; also tells the firmware when to enable I2C_PU.

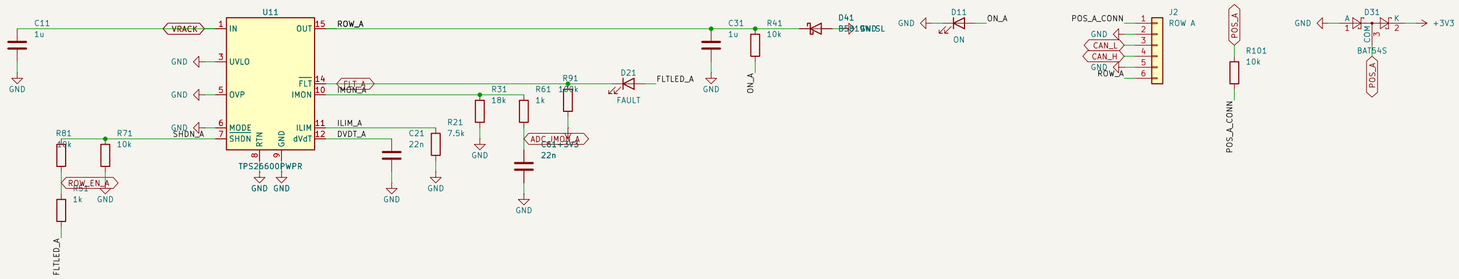
Generated from electronics/power-hub/design.py — edit the Python, then regenerate
USB-C PD distribution + rack controller (ESP32-S3, CAN master) for the hw-testing-rack
hw-testing-rack

Sheet: /Input/
File: input.kicad_sch

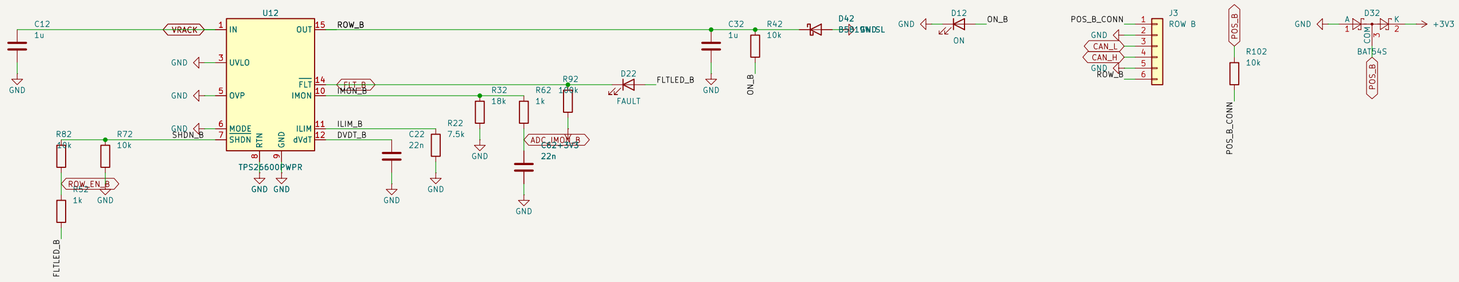
Title: Power hub — USB-C PD input

Size: A3	Date:	**Rev: 0.5**
KiCad E.D.A. 10.0.6		Id: 2/4

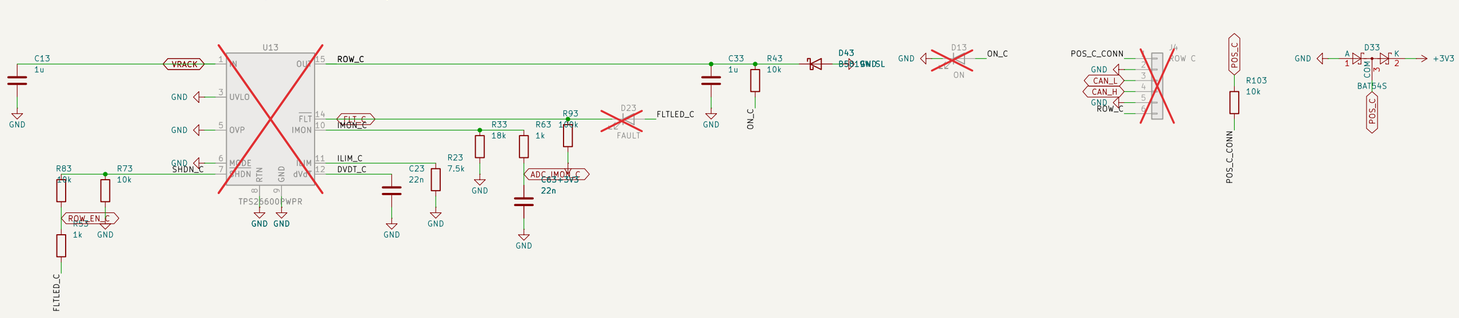
Row A: TPS26600 eFuse 1.6 A, enable, ON / FAULT LEDs, 6-pin JST PH



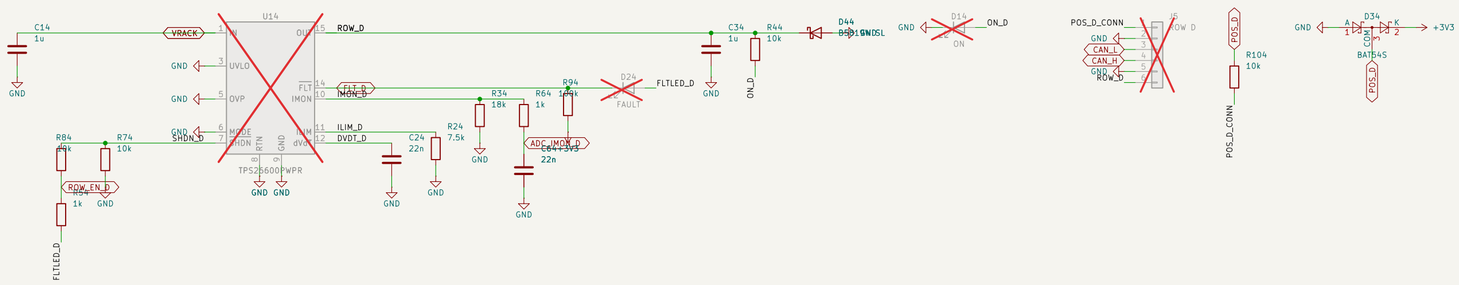
Row B: TPS26600 eFuse 1.6 A, enable, ON / FAULT LEDs, 6-pin JST PH



Row C: TPS26600 eFuse 1.6 A, enable, ON / FAULT LEDs, 6-pin JST PH



Row D: TPS26600 eFuse 1.6 A, enable, ON / FAULT LEDs, 6-pin JST PH



TPS26600: I_LIM ≈ 12 A·kΩ / R_ILIM → 7.5k ≈ 1.6 A (JST PH: 2 A per contact).
t_dVdT = 8e3 · V_IN · C_dVdT → 22 nF ≈ 3.5 ms at 20 V, 4.9 ms at 28 V.
IMON: 78 µA/A into 18k → 1.4 V/A (2.2 V at the limit). 1k + 22 nF to the ESP32's ADC.
SHDN: 10k to GND (row off: ≤ 10 µA from the pin, 0.1 V). ROW_EN through 10k → 1.65 V (on above 0.94 V).
The firmware holds ROW_EN with the ESP32's GPIO hold, so a controller reset leaves the rows as they were.
FLT: open drain, 100k to 3.3 V for the controller; the red LED (1k) is fed from ROW_EN: it lights only for
  a fault on an enabled row (FLT is also low while a row is off or starting).
D41–D44 B5819W: GND → each row's output at its connector, for the cable's kick when the eFuse cuts a short.
UVLO / OVP / MODE to GND: internal 15 V / 33 V thresholds, current-limit mode, auto-retry.

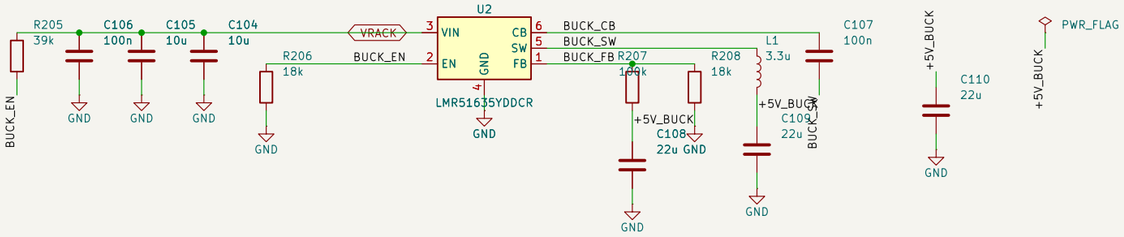
Generated from electronics/power-hub/design.py — edit the Python, then regenerate
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hw-testing-rack

Sheet: /Outputs/
File: outputs.kicad_sch

Title: Power hub — Four eFused row outputs (6-pin JST PH)

Size: A2	Date:	**Rev: 0.5**
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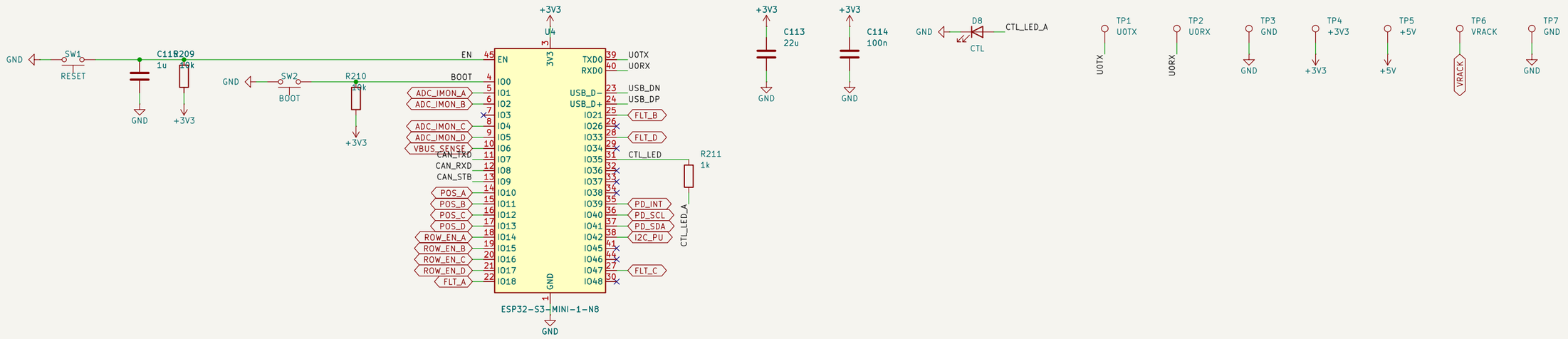
5 V buck: LMR51635, as the bay node (TI SLUSF64B table 8-2), from VBUS ahead of the switch



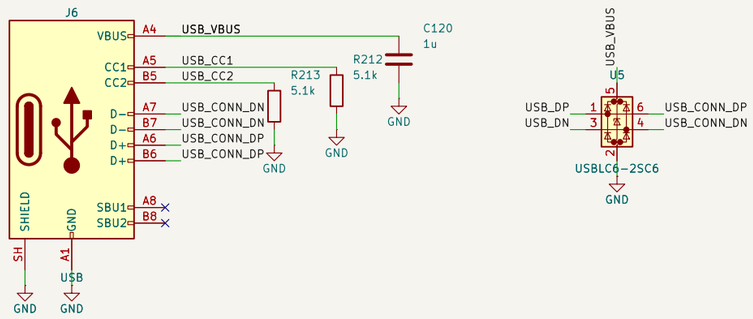
5 V for the controller: the buck or the console USB-C (diode OR), 3.3 V low-dropout LDO



ESP32-S3-MINI-1 (antenna at the board's rear edge: no copper under it, either layer)



USB-C: the controller's console / firmware (native USB), and its 5 V without a PD charger



CAN: TJA1462AT, star bus (all row ports in parallel), both terminations here

Generated from electronics/power-hub/design.py — edit the Python, then regenerate
USB-C PD distribution + rack controller (ESP32-S3, CAN master) for the hw-testing-rack

hw-testing-rack

Sheet: /Controller/
File: controller.kicad_sch

Title: Power hub — Rack controller: supplies, ESP32-S3, USB-C, CAN

Size: A3	Date:	**Rev: 0.5**
KiCad E.D.A. 10.0.6		Id: 4/4